SMART WOUND MONITORING: AI-DRIVEN IMAGE ANALYSIS BASED ON THE T.I.M.E. FRAMEWORK

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UNIVERSITI MALAYA
KUALA LUMPUR

2026

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**RESEARCH REPORT SUBMITTED TO
THE FACULTY OF COMPUTER SCIENCE AND
INFORMATION TECHNOLOGY UNIVERSITI MALAYA,
IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR
THE DEGREE OF MASTER IN DATA SCIENCE**

2026

UNIVERSITI MALAYA

ORIGINAL LITERARY WORK DECLARATION

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Name of Degree: **MASTER IN DATA SCIENCE**

Title of Project Paper/Research Report/Dissertation/Thesis ("this Work"):

SMART WOUND MONITORING: AI-DRIVEN IMAGE ANALYSIS BASED ON THE T.I.M.E. FRAMEWORK

Field of Study: **DEEP LEARNING, AI IN HEALTHCARE**

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ABSTRACT

Chronic wound assessment is a critical yet subjective process in clinical practice. Chronic wounds, such as diabetic foot ulcers, venous ulcers, arterial ulcers, and pressure ulcers, often require continuous monitoring because poor wound management may lead to delayed healing, infection, hospitalization, and reduced quality of life. In routine practice, wound assessment is commonly performed through manual visual inspection and is highly dependent on clinicians' experience and judgement. This may lead to inconsistent interpretation between assessors, especially when evaluating wound tissue, infection signs, moisture level, and wound edge condition. The TIME framework provides a structured guideline for wound assessment by evaluating four key components: Tissue, Infection/Inflammation, Moisture, and Edge. However, its application still remains largely manual. Recent deep learning studies have shown promising results in wound image analysis, especially for wound segmentation and tissue classification. However, many existing approaches focus on isolated tasks and mainly support the Tissue component of the TIME framework. Infection, Moisture, and Edge classification are less commonly integrated into a single automated workflow. In addition, labelled wound images for these tasks are limited, which creates challenges for supervised model training. This study proposes TIME-Net, an automated deep learning framework that integrates tissue segmentation with multi-task classification to support more consistent TIME-based wound assessment. A pre-trained segmentation model, T-SegNet, is used to generate wound masks and produce wound-focused images. These images retain the wound region and nearby periwound area while reducing irrelevant background information. The processed images are then passed into the proposed IME-ClassNet model, which simultaneously predicts Infection, Moisture, and Edge conditions. IME-ClassNet applies transfer learning using EfficientNet-B0 as the shared feature extraction backbone with three task-specific classification heads. To address the limitation of small labelled data, this study applies a semi-supervised pseudo-labelling approach. The supervised baseline model is first trained using the labelled subset and then used to generate pseudo-labels for unlabelled wound images. Only high-confidence predictions are retained and added to the training set, while the validation set remains based on the original labelled data to ensure fair evaluation. Model performance is evaluated using Macro F1-score, sensitivity/recall, and precision. The results showed that segmentation-guided wound-focused images improved IME classification compared with using original wound images directly. EfficientNet-B0 achieved the best overall balance between performance and model efficiency among the tested backbone architectures. The final semi-supervised learning model improved over the supervised baseline, with Macro F1-score gains of 6.83% for Infection, 83.05% for Moisture, and 12.31% for Edge. The final TIME-Net framework was also integrated into a smartphone-based workflow using a cloud-hosted inference pipeline, with a total server inference time of approximately 2.69 seconds. Overall, this study demonstrates the feasibility of combining segmentation, multi-task classification, semi-supervised learning, and mobile integration to support automated wound assessment as a clinical support tool.

Keywords: Chronic Wound Assessment, TIME framework, Deep Learning, Semi-Supervised Learning, Mobile Integration.

ACKNOWLEDGEMENTS

I would like to express my deepest gratitude to all those who have contributed to the success of this project.

First of all, I would like to extend my heartfelt thanks to my project supervisor, Dr. Saw Shier Nee, whose guidance and support have been invaluable throughout this project. Her expertise and mentorship have been instrumental in the progress and completion of this work.

I would also like to acknowledge my collaborator, who provided the T-SegNet model used in this study. The segmentation model played an important role in generating wound masks and wound-focused images, which supported the development and evaluation of the proposed TIME-Net framework.

I am also deeply grateful to my collaborator clinicians, Ms. Saw Shier Khee and her team for their valuable input, domain knowledge, and guidance related to wound assessment and the TIME framework. Their clinical insights helped improve the relevance of this study to real wound assessment practices.

I would also like to express my special thanks to my family and friends for their unwavering support and understanding when undertaking my research. Their encouragement and belief in me have been a source of strength and motivation.

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CHAPTER 1: INTRODUCTION

1.1 Background

The outermost layer of human body is the skin layer. It consists of multiple layers; the epidermis, dermis, and subcutaneous tissue as shown in **Figure 1.1**. Its function is to protect our internal organs and also serves as our first line of defense, protect us against UV rays, diseases, and mechanical harm (Yousef et al., 2024).

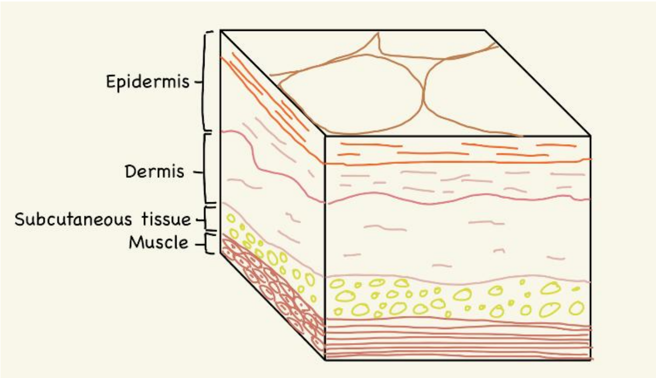


Figure 1.1: Human skin layers, which consist of epidermis, dermis, and subcutaneous tissue (Wehman, 2022)

A wound is formed when the integrity of the skin barrier is compromised by an injury (Afonso et al., 2021). Wounds can be categorized into acute and chronic types based on healing rate and process (Tsegay et al., 2022). Acute wounds usually recover following the normal healing phases, starts from hemostasis, inflammation, proliferation, to remodeling within a predictable timeframe. Whereas, chronic wounds always cannot heal within the expected timeframe (Wallace et al., 2023). Diabetic foot ulcers, venous, arterial, and pressure ulcers are the four types of chronic wounds (Falanga et al., 2022). The treatment plan may differ for each wound type (Boulton, 2020; Robles-Tenorio et al., 2022; Grey et al., 2006; Zaidi & Sharma, 2024).

A study conducted in German reported that the prevalence of chronic wounds was about 1-2% in the general population (Heyer et al., 2016). Some chronic wounds persist for many years and sometimes for decades, which increases the risks of secondary complications such as depression and social isolation. According to the statistical study, the mortality rate after the development of a diabetic foot ulcer was significantly high at 40% (Jupiter et al., 2015). Elderly patients are in higher risk because their skin is thinner and less elastic. Without a proper wound care and management, chronic wounds will easily develop into infection and hospitalization (Shim et al., 2025).

A proper wound assessment is essential to help the wound recover effectively. Clinicians or healthcare professionals need to review some important characteristics of the wound at each review session. These include wound size, tissue type, presence of infection or inflammation, moisture level, and the condition of the wound edge (Mustafa Alhababi et al., 2024). The TIME framework is one of the frequently used assessment tools in wound care management. The 4 key components of TIME are Tissue, Infection or Inflammation, Moisture, and Edge. Tissue component focuses on assessing the wound composition, the non-viable tissue will be removed, while the healthy granulation tissue will be preserved. Infection or Inflammation component addresses bacterial burden. The Moisture component relates to the control of exudate so that the wound environment is not too wet or too dry. The last component, Edge, refers to the epithelial migration from the wound edge. This provides clinicians with a consistent language to describe wound conditions. This allows the development and coordination of effective treatment plans across multiple care settings (Amani et al., 2023).

Digital technology is used more frequently in daily life. According to recent studies, almost two-thirds of people worldwide, including elderly, own a smartphone. From a wound care perspective, clinicians and patients often use smartphones to capture wound photographs as a record. It also enables remote review when frequent clinic visits are difficult. With this, it creates an opportunity to develop a smart wound assessment tool (Blackburn et al., 2019).

1.2 Problem Statement

Wound management is a very important practice in the wound healing process. Patients with chronic diseases need frequent check-ups in order to track their wound healing progress, clean the wound, and get appropriate treatment before it leads to other complications. However, many of these patients have limited mobility and is not convenient for them to travel for a check-up (Amani et al., 2023).

The TIME framework is commonly used as a practical tool for wound care assessment, which focuses on 4 components: Tissue, Infection/Inflammation, Moisture, and Edge of the wound. The traditional wound assessment is purely done manually by healthcare professionals. They usually observe the wound with their naked eyes and record all the observations manually. This can be very time-consuming as each wounds need to be evaluated one by one. This practice also relies heavily on the clinician's experience and different clinicians may have different descriptions and judgements, which can lead to inconsistent decisions about wound dressings and treatments. These inconsistencies can also affect the healing process of the patient (None Rishi Kesaraju, 2025).

With the advancement of imaging tools and mobile devices, wound photographs are now commonly used in clinical practice and research (Chen et al., 2024). Image processing and deep learning techniques also used broadly to automate wound assessment process partially, such as wound segmentation or tissue types annotation. However, most of them are still focusing on isolated tasks, rather than providing full TIME-based wound annotation (Moortgat et al., 2024). This study aims to develop and evaluate a deep learning wound assessment framework that combines a tissue segmentation model and a multi-task IME classifier. Then, apply semi-supervised pseudo-labelling to make use of unlabeled wound images and examine the system's accuracy and practicality for near real-time use in a smartphone-based client application.

1.3 Research Objectives

This research aims to design and evaluate a smart wound assessment framework, known as TIME-Net. This framework consists of 2 deep learning models, which are a tissue segmentation model (T-SegNet) and a multi-task IME classification model (IME-ClassNet).

The objectives of this research are:

1. To develop the IME-ClassNet model, a multi-task deep learning framework for the classification of Infection, Moisture, and Edge of chronic wounds.

2. To evaluate IME-ClassNet performance on a wound image using Macro F1-score and Sensitivity metrics to ensure reliable detection of pathological features.

3. To integrate the IME-ClassNet model, with a tissue segmentation model, T-SegNet, into a smartphone-based prototype, and to evaluate the overall response time.

1.4 Research Questions

To fulfil the research objectives stated above, this study focuses on the following research questions:

1. How effectively does the multi-tasks IME-ClassNet model perform Infection, Moisture, and Edge classification in chronic wound images?

2. What is the classification performance of the proposed IME-ClassNet model in terms of Macro F1-score and Sensitivity metrics?

3. Can the trained TIME-Net, integrating both T-SegNet and IME-ClassNet models, achieve feasible end-to-end response times when deployed as a cloud-based smartphone workflow?

1.5 Scope of Study

The scope of this study is to develop an automated smart wound assessment tool under the TIME framework, known as TIME-Net. This study used a total of 2,831 wound images, consisting of 2,760 unlabeled wound images and 71 labelled wound images annotated for Infection/Inflammation, Moisture, and Edge conditions.

The core analysis is performed by two sub-models: a tissue segmentation model (T-SegNet) and a classification model for Infection, Moisture, and Edge (IME-ClassNet). A pre-trained segmentation model provided by a peer researcher was used to generate the wound masks and produce wound-focused images as the input for IME-ClassNet. The study methodology covers data preparation, model training, testing, and performance evaluation using standard metrics, including Macro F1-score, sensitivity/recall, and precision. The deployment feasibility is evaluated based on the response time of the smartphone-based prototype.

The study also includes the integration of trained models into a mobile application prototype to achieve near-real-time use within a smartphone-based workflow. This focuses on the inference performance and basic display of TIME-related outputs. This study does not include a full clinical trial, long-term outcome tracking, or treatment recommendations. The tool is designed to support, rather than replace, clinical judgement.

1.6 Significance of Study

This study supports more structured and consistent wound assessment in clinical practice. By providing the TIME-based outputs from wound photos, it can provide clinicians with extra objective information instead of relying only on manual visual judgement. Besides, this study also benefits patients with limited mobility. They can assess their wound with just a single capture of their wound using a smartphone at home or in community settings. Overall, this study demonstrates the feasibility of combining wound segmentation, multi-task IME classification, semi-supervised learning, and cloud-based mobile deployment within one TIME-based wound assessment workflow.

1.7 Organization of the Report

This report is organized into 6 chapters. Chapter 1 is the introduction of the study. It consists of the study background, problem statements, research objective, research questions, scope, and significance of the study. Chapter 2 provides the literature review. It discusses factors that affect wound healing for different types of wounds. Then compare existing methods for image-based wound assessment and related work on deep learning, tissue segmentation, IME classification, and mobile-based wound applications. This chapter will also highlight the research gap that this study aims to address. Chapter 3 emphasizes the research methodology. It explains the overall system design of TIME-Net and the main steps carried out to achieve the objective of this research. Chapter 4 presents the results of the study and includes the effect of segmentation on IME classification, backbone architecture comparison, semi-supervised learning results, final model selection, and mobile prototype integration results. Chapter 5 discusses the findings of the study. It interprets the model performance, explains the effect of segmentation and semi-supervised learning, discusses the mobile prototype feasibility, and highlights the limitations of the study. Chapter 6 concludes the report by summarizing the main findings, contributions, limitations, and future work.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Wound care and assessment are important practices in wound management and the healing process, especially for patients with chronic wounds, which take longer than expected time to heal and do not heal quickly on their own (Nagle et al., 2023). Frequent wound check-ups are necessary to prevent the wound getting worst and lead to other complications. Traditional wound assessmnet is done manually and heavily relies on clinician's expertise. Different clinicians may describe the wound differently (Griffa et al., 2024).This can be a major issue as inconsistent description could affect the decision for future treatment plans and subsequently delay the healing progress. The TIME framework provides a structred and standard for wound assessment. It categorizes wound assessment into 4 key components, which are Tissue, Infection/ Inflammation. Moisture, and Edge of the wound. This acts as a guideline for clinicians to assess these four key features of the wound at each review and supports more systematic decision-making. However, TIME-based assessment is still performed manually in most clinical settings, thus, variability still occurs. This limitation should not be ignored because consistent evaluation supports better treatment choices and clearer follow-up on wound healing progress.

Digital imaging has expanded image-based wound assessment in both clinic and home settings. Patient can easily assess their wound with just a single capture of wound images using a smartphone device. With this, they can monitor their wound condition and help in easy documentation anywhere and at any time. This automated wound analysis supports telehealth workflows when in-person review is difficult. Deep learning has shown strong performance in medical image segmentation and classification tasks (LUCAS et al., 2020). This chapter reviews published work on wound assessment, digital and image-based approaches, tissue segmentation and IME classification, deep learning models, integrated frameworks, and mobile deployment, then synthesizes the research gap that motivates TIME-Net.

2.2 Wound Management and Assessment

2.2.1 Wound Classification

There are two types of wounds, which are Acute and Chronic. Wounds are classified based on the severity and healing progress. Wound healing is a very complex biological process that go through 4 overlapping phases: hemostasis, inflammation, proliferation, and remodeling (Wilkinson & Hardman, 2020). No matter acute or chronic wound, both go through the same sequence, but chronic wounds normally takes longer time in inflammation phase, or impaired repair (Schultz et al., 2011).

2.2.2 Factors Affecting Wound Healing

Some key drivers are causing the wound to stall, especially in oxygen-deficient environment. Local hypoxia reduces tissue repair, causing the wound to be stalled in the inflammation phase. Diabetes and vascular disease reduce perfusion and impair immune response and inhibit the tissue regeneration process. Venous hypertension increases oedema and fluid burden. Prolonged pressure reduces blood supply and causes tissue breakdown (Guo & DiPietro, 2010).

A clinical review of wound healing highlighted patient lifestyle and wound characteristics that affect healing results, such as oxygenation, infection burden, diabetes, nutrition supply, smoking behaviors, medication use, etc. (Guo & DiPietro, 2010). According to Aakansha Giri Goswami et al. (2023) and Almuhanna (2025), microbial biofilms cause chronic wounds to heal more slowly by causing persistent inflammation and reduced response to treatment.

2.2.3 Wound Bed Principles

Wound bed preparation principles will be applied based on scenarios, such as removal of necrotic tissues, control the condition of infection, balance exudate level, and support the edge advancement (Schultz et al., 2004). The wound bed preparation is not just to cover the wound, but to prepare the wound bed by correcting all the underlying imbalances (Sibbald et al., 2021).

2.2.4 TIME Framework for Wound Assessment

In early 2000's, Schultz et al, (2003) introduced TIME framework as a practical tool to standardize clinician reviews and to link findings to practical actions. It focused on 4 specific barriers to healing, which are Tissue, Infection/ Inflammation, Moisture, and Edge of wound. Each component requires specific visual observations that determine the next treatment plan (Schultz et al., 2003). The first component, Tissue, focuses on the wound bed composition, which can be classified into necrotic tissue, slough, and granulation. It acts as an indicator to help clinicians differentiate between viable (healthy red granulation tissue) and non-viable tissue (yellow slough and black necrotic tissue). Debridement is required when the non-viable tissues are identified, as it will block the healing process (Garten et al., 2020). For Infection of Inflammation, it focuses on bacterial burden or host response. The classic signs and symptoms can be summarized as NERDS (Non healing, exudate increase, red friable tissue, debris, and smell), surrounding redness, purulence, or cellulitis. Moisture balance covers the wound fluid environment (Orsted et al., 2018). Excess exudate and periwound maceration indicate too wet condition. Whereas dryness and adherent tissue suggest insufficient moisture. The class between these two is considered moderate moisture, which is the best condition for the wound to be repaired. The fourth criterion is the edge of the wound, which focuses on epithelial advancement and margin condition. Conditions like rolled edges, callus, and poor periwound, are an indication for stop healing (The Royal Children's Hospital Melbourne, 2023).

2.2.5 Challenges with TIME Framework

Other than the standardization of TIME framework, its application in routine clinical practice remains heavily dependent on subjective visual inspection. For example, measurement of wound size is often done by using a ruler, tracing, or digital planimetry (A Comparison of Computer-Assisted and Manual Wound Size Measurement, 2026). Besides, the classification of tissue type, sign of infection, exudate level, or edge advancement is all observed using the naked eye. Studies such as Vestjens et al. (2017), have shown that different clinicians frequently disagree on tissue classification. They got low agreement between nurses when determining the percentage of slough over granulation tissue of a wound. This further support the need of a more objective and repeatable assessment tool, especially when the consistent reporting is required under structured TIME framework.

2.3 Digital and Image-Based Wound Assessment

Wound photographs is the main component for digital wound assessment in order to support documentation, follow-up, and remote reviews. McLean et al. (2023) study showed that remote digital wound monitoring tool has been implemented in 2 UK Hospitals. They prove that remote wound assessment is applicable within a real care pathway. Similar findings from Gunter et al. (2016) also demonstrate that high usability and acceptance of smartphone-based wound monitoring when patients are guided for image capture and reporting. These approaches successfully reduce the need of frequent in-person visits and help track wound status across time.

Image quality is the key requirement for any image-based assessment. If the image is poor in quality, such as insufficient lighting, incorrect angle, distance, blurriness, and the absence of a scaling reference object, it may affect the annotation results from both human and automated analysis. Zhang et al. (2021) developed a home-based mobile health tool that included real-time feedback to help patients capture usable wound images, and they reported that this feedback improved overall image quality while supporting home monitoring. This prove that structured capture guidance is important and needed for a digital workflow to provide accurate and reliable assessment.

Digital wound assessment also includes measurement and tracking tools. Digital methods aim to improve consistency in wound size and feature recording compared with manual measurement. A systematic review evaluated digital devices for wound measurement and highlighted ongoing variation in accuracy and reporting across tools, which shows the need for careful validation and standard measurement practice (Kientzy et al., 2024). In the same direction, published work continues to propose practical solutions to reduce measurement error from photo-taking variation. They have added reference markers to support consistent scaling (Li et al., 2025). These studies support the value of digital methods, while also showing that capture conditions and standardisation still shape reliability.

Automated features are currently available in a large number of commercial and academic applications; however, the strength and quality of the data differ between applications. Kabir et al. (2024) evaluated wound assessment apps and discovered gaps in the evidence-based reliability of existing apps. Another evaluation of AI-integrated mobile apps for ulcer segmentation reveals that many systems still focus on a particular task, primarily segmentation, rather than a complete clinical framework (Griffa et al., 2024). This creates a need for existing research that links multi-component evaluation, segmentation, and image capture under a structured model like TIME with performance and usability evaluated for mobile devices.

2.4 Tissue Segmentation and IME Classification Pipeline

2.4.1 Tissue Segmentation

Tissue segmentation is always the start for image-based wound assessment. This is because the model needs to first separate the wound from the surrounding skin and background. This step is important as a wound-focused input can reduce noise from any irrelevant background objects (Veredas et al., 2015). Previous studies used traditional image processing methods such as thresholding, clustering, and contour-based techniques to show the wound boundaries (Anisuzzaman et al., 2021). However, these methods often perform poorly when image conditions vary, such as lighting, skin tone, camera angle, or wound appearance (Ahmad Fauzi et al., 2015).

Deep learning segmentation reduces reliance on hand-made rules by learning wound features directly from labelled images. Many studies use encoder-decoder designs, often based on U-Net, to produce pixel-level wound masks. After that, they also evaluate the result using IoU or Dice (Cassidy et al., 2021). A challenge report on diabetic foot ulcer segmentation also describes the evaluation of localization, classification, and segmentation methods under shared rules, which shows that wound segmentation has matured into a benchmarked task (Yap et al., 2024).

In this study, tissue segmentation was used as an upstream step to generate wound masks and produce wound-focused images. Instead of training the segmentation model as the main contribution, this study used the segmentation output to reduce background noise and guide the IME classification stage. The wound region was retained while the surrounding non-wound background was masked, with padding included around the wound region to preserve useful edge and periwound information.

2.4.2 IME Classification

The IME classification covers infection or inflammation, moisture balance, and edge of wound. Infection or Inflammation classification often receives more attention because infection links to risk and escalation. A review of machine learning approaches for image-based identification of surgical wound infections summarized published image-based methods and stressed the need for stronger validation across settings (Pablo et al., 2024). For moisture and edge, literature often treats them indirectly through tissue and periwound cues, or through segmentation outputs, because these features can be subtle and image quality can change them. Chang et al. proposed a super pixels approach to help tissue labelling for classification, which enhances tissue feature extraction from wound images. Besides, Chairat et al. (2023) demonstrated that smartphone images can allow multi-region outputs when labels are provided by using deep learning methods to segment wound area and tissue regions.

2.4.3 Integration of Segmentation and Classification

In medical image analysis, segmentation and classification are often independent tasks. Segmentation isolates the wound region, also known as the region of interest, while classification assigns a diagnostic label to that region. There are two architectural strategies that can be used to integrate both segmentation and classification simultaneously, which are a sequential two-stage pipeline or a unified multi-task network.

In sequential two-stage pipeline, segmentation will first isolate the region of interest, then classification will predict labels from the segmented region. A breast ultrasound study described a "segmentation-guided classification" pipeline, where lesion segmentation acts as the input for the classification stage for diagnosis, showing the two steps work together as one decision workflow (Inan et al., 2022). Similar two-stage designs also appear in broader medical imaging, where a first stage localizes or segments the target and a second stage performs diagnosis. This is very helpful, especially when background dominates the image, as this can cause bias classification (Abdel-Basset et al., 2021).

Another design is the multi-task network, which merges segmentation and classification in a single model. In this approach, one network shares early feature layers and produces both a pixel-level mask and a classification output. A dual-branch skin lesion study reported a framework that performs segmentation and classification together, which supports unified training and consistent feature learning for both outputs (Owida et al., 2025). This model does not require two models to run, but it requires careful training because both outputs must learn well at the same time.

2.4.4 Address Label Quality and Variability

Label quality is a key limitation for both segmentation and IME classification. Image labels often come from visual judgement, and agreement is not equal across tissue types. A deep learning study on wound tissue segmentation reported inter- and intra-rater variability during annotation and treated this as a practical barrier for objective assessment. This is important for IME tasks because infection, moisture, and edge labeling all rely on visual indicators that are influenced by experience and lighting (Ramachandram et al., 2022). These results support the study's approach, which focuses IME characteristics on the wound region using a segmentation output and assesses performance using standard metrics for both segmentation and classification.

2.4.5 Evidence from Wound-Specific Studies

Multi-task pipelines reduce duplication and can share useful features, but they need good labels for both tasks and careful balancing of losses during training. Wound-specific evidence also supports combined workflows. A pilot study trained deep learning models for wound segmentation and wound classification on the same image set, showing that both tasks can be handled within one study design, while also reflecting the practical need for reliable segmentation and labels when classification depends on wound region cues (Zimmermann et al., 2023).

2.5 Deep Learning Models for Medical Image Classification

The limitation of traditional image processing has brought the adoption of Deep Learning. Convolutional neural networks (CNNs) are a primary model family use for many medical image tasks, such as segmentation, detection, and classification (Litjens et al., 2017; Cai et al., 2020). They also summarize common architectures such as AlexNet, VGG, GoogleNet, and ResNet. These models often appear in 2 practical ways. First way is to train a network from scratch when a large, labelled dataset exists. The second way is transfer learning, which starts from a pre-trained model on a large image dataset, then transfer the knowledge to the smaller dataset (Esteva et al., 2019). This technique is more frequently applied in medical dataset, as medical data usually are small and limited (Shen et al., 2017).

Many medical problems also involve more than one label or more than one related task. Multi-task deep learning addresses this by learning shared features for related outputs, then producing separate predictions for each task. This technique align with our IME model, this model also using a shared feature extractor and covers 3 wound features. In practice, multi-task models often share early layers for general visual features, then branch into task-specific heads near the end of the network. The goal is improved learning efficiency and better generalisation when tasks share visual cues (Shim et al., 2025).

Model evaluation in medical image classification relies on the metrics that match the specific clinical goal and the class balance of the dataset. Accuracy alone can hide poor performance on minority classes when classes are imbalanced. Therefore, matrices like precision, recall, and F1-score are often reported because they describe different types of errors and reflect the false positives and false negatives (Litjens et al., 2017). Macro-averaged scores also help reflect performance across all classes rather than just bias towards the domain class. Confusion matrices also help explain which categories are most often confused, which supports error analysis and label refinement (Garcia et al., 2015).

2.6 Semi-Supervised Learning for Limited Labeled Data

Small dataset and class imbalance are the two commonly seen challenges in medical image classification studies. A small dataset will increase overfitting risk, especially when image capture conditions are different across hospitals or devices. Class imbalance occurs when severe cases occur less often than normal cases. This can lead to biased towards the majority classes when training images are different from deployment images in terms of lighting, camera type, patient group, or clinical settings (Cheplygina et al., 2019). Transfer learning, data augmentation, class-weighted losses, resampling techniques, and strong validation designs are some popular approaches (Morid et al., 2021).

Semi-supervised learning also provides a solution by increasing the volume of data by using both labelled and unlabeled datasets (Liu et al., 2023). One common approach is pseudo-labelling, where a model is first trained using the available labelled dataset and then used to generate predicted labels for unlabelled images. Predictions with high confidence are retained as pseudo-labels and added to the training set. This allows the model to learn from a larger variety of images without requiring all images to be manually annotated.Reviews also emphasize the importance of clearly describe and evaluate data from many sources in order to show generalization (Zech et al., 2018).

2.7 Mobile Deployment of Medical Analysis Models

Mobile deployment refers to running medical image analysis within a smartphone-based workflow, where the smartphone is used to capture the image. There are two ways to execute model inference, which are on-device (edge inference) and remotely (cloud-hosted inference). Edge inference supports faster feedback because it reduces reliance on uploading images to a remote server. However, studies on mobile deep learning show that deployment on smartphones faces practical limits in terms of processing power, memory, battery use, and heat, so model design must strike a balance between accuracy, speed, and resource consumption (Wang et al., 2021). Therefore, studies often choose to use lighter model families and apply some optimization including the use of compact backbones, reduce input resolution, and apply compression to shrink the model size in order to speed up the inference time. A wound assessment tool should return results fast for routine treatment and home monitoring, but the input resolution cannot be reduced, as it will affect the result of wound annotation.

On the other hand, because cloud-hosted inference removes the need for runtime integration and mobile-specific model conversion, it is commonly used in prototype-stage systems. This approach relies on a smartphone application to send the input wound images (or extracted regions of interest) to a managed inference endpoint, which then uses an API to return segmentation and classification results. This design supports rapid iteration and centralized updates of model versions.

In this study, the TIME-based pipeline was integrated into a smartphone-based prototype that performs inference through a managed cloud inference service accessed via API. The wound image is uploaded from the mobile application to Hugging Face Spaces, which acts as the cloud inference endpoint. The image is processed through the TIME-Net pipeline, including T-SegNet, preprocessing, and IME-ClassNet, before the prediction results are returned to the Flutter application and displayed in the Android Studio emulator. This deployment approach supports rapid prototyping and avoids direct on-device model conversion, while allowing the response time of the full workflow to be evaluated.

2.8 Research Gap

Previous studies confirm that deep learning can support wound image analysis, especially for wound segmentation and tissue-type classification. This progress supports more objective wound measurement and improves consistency compared to manual justification. The TIME framework is widely used in clinical wound care, but it is still performed manually and depends on visual judgement. This can lead to subjectivity and variability, which will also affect consistency in treatment decisions.

There are three gaps that remain to be filled in this study. Firstly, current work lacks holistic TIME-based assessment. Most deep learning studies focus on isolated tasks, mainly wound segmentation or tissue classification, which addresses the “Tissue” component of TIME. There are lacking of studies focus on Infection or Inflammation, Moisture, and Edge of the wound within a TIME-based assessment workflow. Current tools do not provide complete multi-task model that reflect routine TIME-based clinical review (Morgado et al., 2025).

Secondly, there are lacking multi-task models that focusing on all 3 Infection/ Inflammation, Moisture, and Edge (IME) features. Segmentation and clinical feature prediction are always treated as separate steps or implemented as separate models. This increases computational cost and system complexity. There are still a need for an integrated pipeline where segmentation focuses the region of interest and a shared-feature multi-task classifier predicts the IME outputs together, using consistent inputs (Mohammed et al., 2025).

Thirdly, deployment feasibility remains unclear for complete TIME-style pipelines. Smartphone-based wound photography is common, but many deep learning models are developed and evaluated in desktop or server environments. Few studies report end-to-end inference performance for combined segmentation and multi-attribute IME classification on standard smartphone devices. These limits understanding of whether such systems can achieve near-real-time inference particularly when inference is provided through managed cloud services accessed via APIs (Anisuzzaman et al., 2021).

Therefore, this study addresses these gaps by proposing TIME-Net, an integrated deep learning framework that combines a tissue segmentation model (T-SegNet) with a multi-task IME classification model (IME-ClassNet). Then, evaluate the analytical performance and deployment feasibility to support structured, objective, and accessible TIME-based wound assessment.

CHAPTER 3: METHODOLOGY

3.1 Overview of Proposed Framework

This study proposes an automated wound assessment framework known as TIME-Net. It was designed to support wound assessment based on the TIME framework using wound images captured by smartphone devices. The main purpose of developing TIME-Net is to help provide a more structured and consistent evaluation of chronic wounds by analyzing key features related to Tissue, Infection or Inflammation, Moisture, and Edge conditions.

TIME-Net is the workflow that consists of 2 main components: a tissue segmentation model, T-SegNet, and a multi-task classification model, known as IME-ClassNet. The T-SegNet functions to segment out the wound area and provide a wound mask that helps the analysis focus on the wound itself and reduce the background noise. The segmented wound images will then be passed to the IME-ClassNet model to predict the Infection or Inflammation, Moisture, and Edge condition at the same time.

In this study, the T-SegNet model was developed by a collaborator and was used as a pre-trained model. It will only used to generate the wound masks as the input for IME-ClassNet model. The design, training, and optimization of T-SegNet are not part of the scope of this study. Instead, this research fully focuses on how the segmented output can be used to support more accurate and consistent IME classification.

The main contribution of this study still lies in the development and evaluation of IME-ClassNet, and also the integration of both models into a single TIME-based assessment framework that can be used on a smartphone device. The main goal of this study was to achieve reliable IME classification performance using limited labelled data and to demonstrate that the overall TIME-Net workflow can produce results within a short response time suitable for a smartphone-based prototype. The overall TIME-Net workflow, from wound image input to TIME-based result output is shown in **Figure 3.1**.

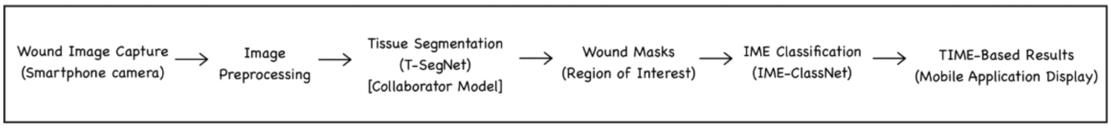


Figure 3.1: Overview of the proposed TIME-Net Framework

3.2 Dataset Description

This study used a publicly available wound image dataset obtained from Kaggle, which contains 2831 wound images compiled from multiple sources (Thomas, 2020; Wang et al., 2020; Subba Reddy Oota et al., 2023). These images include different wound appearances, lighting conditions, backgrounds, and angles.

This dataset only consists of limited subset that had labels for Infection/Inflammation, Moisture, and Edge based on the TIME framework. These labelled images were used for supervised training and validation of IME-ClassNet, while the remaining unlabeled images were used for semi-supervised pseudo-labelling. As the labelled subset showed class imbalance across IME tasks, especially Moisture, class-weighting loss was applied during model training and Macro F1-score was selected as the main evaluation metric.

3.3 Data Preprocessing

All wound images underwent a series of preprocessing steps to ensure consistency and to readily prepare for analysis. This step helps to reduce variation caused by image capture conditions and to allow fair comparison during model training.

A brief manual quality check was performed to remove images with severe quality issues, such as extreme blur or incomplete wound visibility. Some of the wounds have multiple images from different angles. All the images were grouped by independent wound, which helps to prevent data leakage when both images are assigned to both the training and validation groups later.

3.3.1 Region of Interest (ROI) Extraction

Next, the wound images were passed through the tissue segmentation model (T-SegNet) to generate the wound masks. The generated masks were then overlapped with the original RGB images to produce wound-focused images. In the processed output, the wound region was retained in RGB colour, while the surrounding non-wound area was masked in black. Padding was also added around the wound region to preserve the wound edge and nearby periwound area, as these regions may contain useful visual cues for Infection, Moisture, and Edge classification.

3.3.2 Image Standardization

After wound-focused image generation, all images were resized to 224 x 224 pixels as required by the IME-ClassNet model. Bilinear interpolation was used to resize images because it provides smooth image scaling without having sharp artefacts. Pixel values were then normalized to the range [0, 1] by dividing by 255 to reduce the effect of lighting differences and to support efficient model learning. The color information was preserved, and images were processed in RGB format. This was because color cues are important for identifying Infection or Inflammation, Moisture, and Edge condition of a wound.

3.4 Exploratory Data Analysis (EDA)

EDA was performed to gain a basic understanding of the data and guide modelling decisions.

3.4.1 IME Class Distribution

For the labeled subset, EDA was performed to examine the distribution of IME labels. As shown in **Figure 3.2**, the dataset shows significant class imbalance across all 3 features.

In particular, the graph for the Moisture task skewed towards ‘Moisture’ class (N = 46), which dominates the minority classes (‘Too_Dry’ and ‘Too_Wet’). The class distribution patterns observed in **Figure 3.2** influenced the selection of class-weighted loss during training and Macro F1-score as the main evaluation metric.

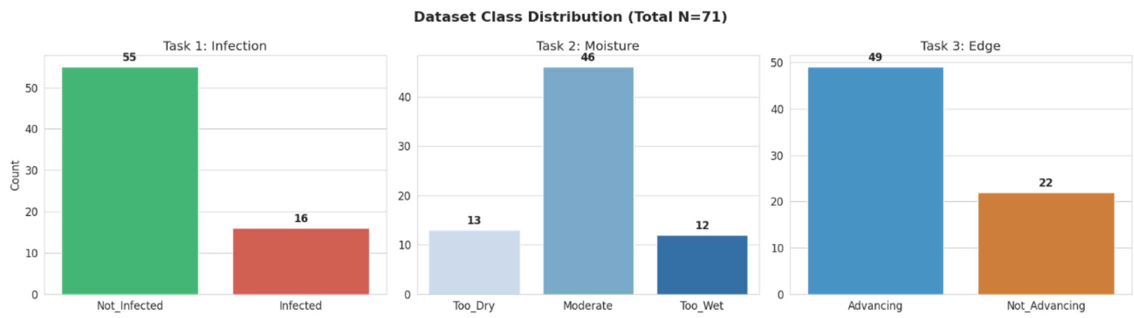


Figure 3.2: Class Distribution for Infection, Moisture, and Edge tasks in the labeled subset (N = 71).

3.4.2 Image Quality and Variability

EDA was also used to assess variation in image quality and capture conditions, such as differences in lighting, background, distance, and viewing angle. This inspection helped confirm the need for preprocessing steps such as normalization, ROI extraction, and data augmentation during model training. Images with poor quality were excluded through the quality filtering process as described in Section 3.3.

3.5 Tissue Segmentation (T-SegNet)

Tissue segmentation is an important step in the proposed TIME-Net framework. It functions to identify the wound region accurately from the wound images. In this study, tissue segmentation was used to support the IME classification by locating the wound area and guiding the ROI extraction.

3.5.1 Role of T-SegNet

The tissue segmentation model used in this study is known as T-SegNet. It was developed by a collaborator and provided as a pre-trained model. The input to T-SegNet was the RGB wound image, and the output was the wound mask used to identify the wound region. For the purpose of this study, the segmented wound mask helps to distinguish the wound area from the irrelevant background.

3.5.2 Integration and Scope

The T-SegNet model was treated as a fixed upstream component that produces binary wound masks. The internal architecture, training process, and optimization of T-SegNet are not part of this research. The segmentation output was only used to produce wound-focused images through mask overlay and padding. These processed images were then used as the input for IME-ClassNet.

3.6 IME-ClassNet Model Architecture

The main contribution of this study was the IME-ClassNet, which is a unified deep Convolutional Neural Network (CNN) designed to predict the classification of the 3 wound features: Infection, Moisture, and Edge condition. A transfer learning approach was adopted by utilizing a Multi-Task Learning (MTL) architecture based on a lightweight convolutional backbone. The goal was to achieve high inference efficiency within a smartphone-based wound assessment workflow with accurate diagnostics.

3.6.1 Multi-Task Learning Design

IME-ClassNet uses a shared feature extraction approach, where a single backbone network learns common visual features and branches into separate task-specific output heads. There are 2 benefits of this design. It helps improve computational efficiency, as all 3 IME classification tasks are predicted in a single forward pass, to reduce memory usage and inference time, which is important for smartphone-based workflows with cloud-hosted inference. Next, the model captured shared information and learned these tasks simultaneously as the IME attributes were clinically related and often share the same signs. This can help to reduce overfitting risk, especially when there is limited data.

3.6.2 Convolutional Backbone Network

The shared feature extractor in IME-ClassNet was based on the EfficientNet-B0 architecture. EfficientNet-B0 processed the preprocessed wound image data through convolutional layers to extract visible features, which are related to color, texture, and shape. Global average pooling approach is used to condense the final feature map into a compact feature representation. This shared feature representation was then used as the input for the task-specific classification branches.

3.6.3 Task-Specific Classification Heads

IME-ClassNet includes three independent classification heads, each corresponding to one component of the IME framework. The inflammation and edge heads perform binary classification, where the Infection head predicts infected or not infected, and the Edge head predicts advancing or not advancing. At the same time, the moisture head performs multi-class classification to categorize the wound exudate level into too dry, moderate, and too wet.

While each classification head generates its own task-specific output, they all receive the same shared feature representation. Each wound image's final IME assessment was generated using the probability scores from these outputs.

3.7 Model Training

The IME-ClassNet model was trained using the preprocessed wound images. The model training pipeline was designed to make effective use of the limited labeled dataset while reducing the risk of overfitting and bias towards majority classes.

3.7.1 Supervised Transfer Learning and Fine-Tuning

Transfer learning approach was applied to the IME-ClassNet model to improve training efficiency under limited labelled data conditions. The EfficientNet-B0, which was pre-trained on the ImageNet dataset, was selected to serve as the backbone network. This allows the model to learn basic feature detectors such as edges, textures, and color patterns, as the starting point for wound analysis.

The core backbone of EfficientNet-B0 was frozen during the first training phase, and only the task-specific classification heads were trained. This allows the model to adapt the IME classification tasks without changing its general visual representation. After that, the top layers of the EfficientNet-B0 backbone were gradually “unfrozen” and fine-tuned using a lower learning rate to allow further adaptation to wound-specific features.

3.7.2 Data Augmentation

The training set was applied with data augmentation techniques such as random rotations, flipping, zooms, cropping, and lighting adjustments to increase the size for training, which also replicates the real-world variations in camera angles, to achieve model generalizability (None Rishi Kesaraju, 2025).

3.7.3 Handling Class Imbalance

The class-weighted loss technique was used to reduce the bias of the model towards majority class. The loss weights were assigned according to the relative frequency of each IME category. With this, the model was encouraged to gain balanced representations of all components, including conditions that appear less frequent.

3.7.4 Optimization Configuration

Model optimization was performed using the Adam optimizer. Training was performed in small batches to manage memory usage efficiently and stopped when validation performance no longer improved.

3.8 Semi-Supervised Learning

Semi-supervised learning approach based on a pseudo-labelling approach was proposed to overcome the limitations of the small labelled subset. This method used the labelled subset to train an initial supervised baseline model, and then used the trained model to generate pseudo-labels for the unlabelled wound images. High-confidence pseudo-labelled images were added to the training set to increase the amount of training data without requiring manual annotation (Liu et al., 2023).

3.8.1 Baseline Model Training

A baseline IME-ClassNet model was trained using the labelled subset. This training follows the procedure in Section 3.7. Although this baseline model was trained on a small labeled subset, it provided an initial IME classifier that captured the fundamental characteristics of IME from wound images. This served as the starting point for generating predictions on unlabeled data in the next phase.

3.8.2 Pseudo-Label Generation and Filtering

The trained baseline IME-ClassNet model was used to predict Infection, Moisture, and Edge labels for the unlabelled wound images. Each prediction was generated together with a confidence score. A confidence-based filtering strategy was then applied, where only predictions that achieved the selected confidence threshold were retained as pseudo-labels.

The retained pseudo-labelled images were added only to the training set, while the validation set remained based on the original labelled data. This was done to ensure a fair evaluation and to prevent pseudo-labelled data from influencing the validation results. Different confidence thresholds were tested across several pseudo-labelling rounds, and the best-performing model from each round was selected for the next round.

3.8.3 Model Retraining

A new instance of the IME-ClassNet model was retrained using the expanded training set that includes both original labeled images and the selected pseudo-labeled images. This newly combined dataset was used to train a new model, which starts with pre-trained ImageNet weights and training using the same procedure as described in Section 3.7.

The new model was exposed to more wound appearances, lighting conditions, and image capturing angles, which subsequently improve the model generalizability. This pseudo-labelling process was repeated across several rounds until the model performance reached a plateau.

3.9 Model Evaluation

The IME-ClassNet model performance was evaluated using the metrics below:

3.9.1 Stratified k-fold Cross-validation

As this study has limited labelled wound images, a traditional train-test split is not appropriate to be used in this study. On the other hand, this study used a stratified k-fold cross-validation method to test model performance. This method allowed all the images used for both training and validation across different folds.

In this study, the newly labeled dataset was divided into 5 folds while maintaining the distribution of IME labels across folds. For each round, one fold was used for validation, and the other 4 folds were used for training. This process was repeated until every fold had been used once for validation, and the average across 5 folds was reported as the final model performance.

3.9.2 Evaluation Metrics

IME-ClassNet model performance was evaluated using standard classification metrics such as Macro F1-score, Sensitivity (Recall), and Precision. The standard accuracy was not used as the primary metric because of the class imbalance of the 3 features.

Macro F1-score was selected as the main evaluation metric because it gives equal importance to each class and provides a more balanced view of performance compared to accuracy alone.

$$\mathrm{F1-Score} = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Sensitivity measures the ability to identify positive cases, such as infection or not advancing edges.

$$\mathrm{Sensitivity} = \frac{TP}{TP + FN}$$

Precision measures the reliability of positive predictions. This helps to check for the rate of False predictions.

$$\mathrm{Precision} = \frac{TP}{TP + FP}$$

3.9.3 Reporting

For each cross-validation fold, evaluation metrics were calculated on the validation data. Final model performance was reported as the mean and standard deviation across all folds to reflect both accuracy and stability. Confusion matrices were used to analyse common misclassification patterns across the IME features.

3.10 Model Deployment and System Evaluation

The proposed TIME-Net framework was then integrated into a smartphone-based prototype workflow for near real-time wound assessment.

3.10.1 Model Deployment Framework

The trained IME-ClassNet together with the T-SegNet model, was combined as TIME-Net and integrated into a smartphone-based prototype. The inference workflow was hosted using Hugging Face Spaces, which acted as the cloud inference endpoint. The smartphone-based prototype was developed using Flutter and Dart, and the application was tested using the Android Studio emulator.

In the deployment workflow, the wound image was uploaded from the mobile application to the cloud inference endpoint. The image was then processed through the TIME-Net pipeline, including tissue segmentation, preprocessing, and IME classification. The prediction results were returned to the Flutter application and displayed to the user as TIME-based wound assessment outputs.

3.10.2 End-to-End Performance Benchmarking

In the prototype, inference was performed through a managed cloud inference service (remote inference), while the smartphone device functions as the image capture and result display interface. This design simplifies deployment and careful handling of wound images to address privacy and governance considerations. Performance was evaluated using end-to-end response time, measured from the moment an image is submitted until the full TIME outputs (mask + IME predictions) were returned and displayed, including network transmission and server-side inference time. Where applicable, the total response time can be reported as the sum of (i) T-SegNet inference time and (ii) IME-ClassNet inference time.

3.10.3 Performance Target and Evaluation Criteria

End-to-end response time was measured and analyzed to assess the deployment feasibility of the proposed cloud-based TIME-Net framework, which was considered acceptable for practical use in clinical or home-based wound monitoring scenarios.

Evaluation was performed by repeating inference runs across multiple wound images and reporting the mean response time.

3.11 Summary

This chapter has presented the methodology used for developing and evaluating the proposed TIME-Net framework for image-based chronic wound assessment. All the wound images underwent data preparation, including image standardization, region-of-interest extraction guided by tissue segmentation, and quality filtering. A pre-trained tissue segmentation (T-SegNet) developed by collaborator was used to support the IME classification prediction. The IME-ClassNet architecture was designed as a multi-task learning with lightweight pre-trained backbone, which suitable for mobile deployment.

Transfer learning, data augmentation, handling class imbalance, and optimization considerations were included in the model training section. A semi-supervised learning approach based on pseudo-labelling was introduced to overcome the small dataset limitation. The model was evaluated using stratified 5-fold cross-validation and classification metrics such as Macro F1-score, sensitivity/recall, and precision. Finally, the selected model was integrated into a smartphone-based prototype using a cloud-hosted inference workflow through Hugging Face Spaces and a Flutter-based mobile application. The system feasibility was evaluated based on response time and successful display of TIME-based wound assessment results.

CHAPTER 4: RESULTS

4.1 Effect of Segmentation on IME Classification

A preliminary test was conducted to determine the effect of segmentation on IME classification performance using raw wound images and segmented wound images.

Table 4.1: Performance comparison with and without segmentation

Approach	Infection Macro F1	Moisture Macro F1	Edge Macro F1	Average Macro F1
Without Segmentation	0.5564	0.3407	0.5509	0.4827
With Segmentation	0.6929	0.3576	0.7403	0.5969

Table 4.1 showed the segmented inputs have better IME performance across all 3 tasks compared with the original wound images. The average Macro F1-score increased from 0.4827 to 0.5969 after segmentation was applied. The largest improvement was the Edge classification, followed by Infection classification. This suggests that segmentation helped the model focus on wound-related regions and reduce irrelevant background information. Therefore, the segmentation approach was used for the following experiments.

4.2 Backbone Architecture Comparison

There were four architectures compared under the same multi-task classification setting, which were EfficientNet-B0, ResNet50, MobileNet_v3, and Swin_t. In this comparison, the main metrics were Macro F1-score for each task and the model size.

Table 4.2: Backbone architecture comparison for IME-ClassNet

Approach	Parameters (M)	Infection Macro F1	Moisture Macro F1	Edge Macro F1	Average Macro F1
EfficientNet-B0	5.3	0.6929	0.3576	0.7403	0.5969
ResNet50	25.6	0.6281	0.4409	0.5913	0.5534
MobileNet_v3	5.4	0.5482	0.3655	0.5653	0.4930
Swin_t	28.3	0.5948	0.4326	0.6304	0.5526

Based on Table 4.2, EfficientNet-B0 showed the highest average Macro F1-score of 0.5969. It also achieved the best performance for Infection and Edge classification. Although ResNet50 achieved the highest Moisture Macro F1-score, its average Macro F1-score was lower than EfficientNet-B0 and it had a much larger number of parameters. Therefore, EfficientNet-B0 architecture was selected as the backbone for the following experiments.

4.3 Semi-Supervised Learning and Final Model Selection

After that, semi-supervised learning was conducted using pseudo-labelling for 4 rounds. In each round, different confidence thresholds were tested, and the best-performing configuration was selected for the next round. Round 1 to 4 models were first compared using checkpoints selected based on the best total validation loss.

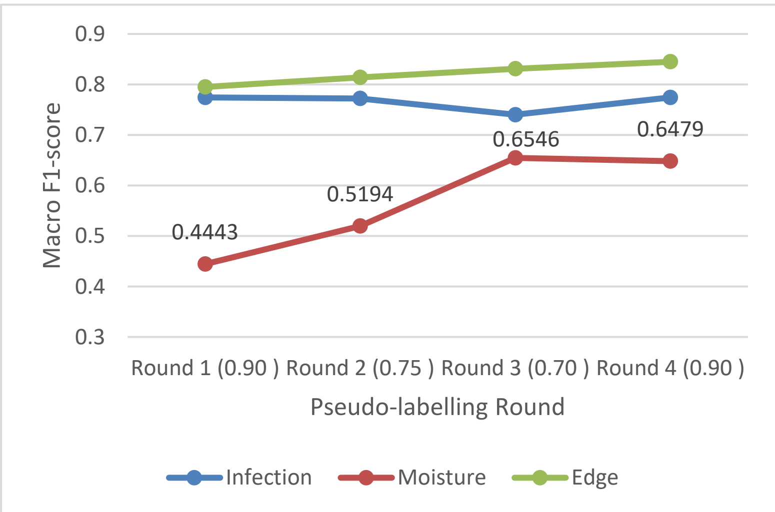


Figure 4.1: Macro F1-score across semi-supervised pseudo-labelling rounds

Based on Figure 4.1, the Edge task achieved the most stable and highest Macro F1-score across all pseudo-labelling rounds. Its performance increased gradually from Round 1 to Round 4. The Infection task showed relatively stable performance, with a slight drop in Round 3 before increasing again in Round 4. In comparison, the Moisture task showed the most noticeable improvement across the pseudo-labelling rounds. The Moisture Macro F1-score increased from 0.4443 in Round 1 to 0.5194 in Round 2 and reached 0.6546 in Round 3. This suggests that pseudo-labelling helped the model learn additional moisture-related visual features from the unlabeled wound images.

However, the Moisture Macro F1-score slightly decreased from 0.6546 in Round 3 to 0.6479 in Round 4. Although Round 4 generated more high-confidence pseudo-labels than Round 3, it did not achieve the best Moisture performance. The active Moisture pseudo-labels increased from 530 in Round 3 to 2243 in Round 4, but the best Moisture Macro F1-score was still achieved by the Round 3 model. This shows that increasing the number of pseudo-labels does not always improve model performance, as pseudo-label quality is more important than pseudo-label quantity.

Table 4.3: Performance comparison between the supervised baseline and final proposed SSL model

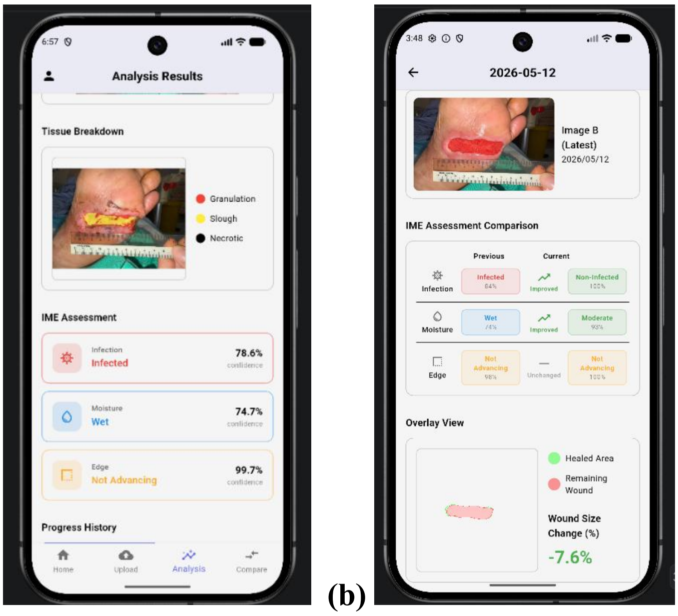
Task	Supervised Baseline	Proposed SSL Model	Absolute Gain	Relative Improvement (%)
Infection Macro F1-score	0.6929	0.7402	+0.0473	6.83%
Moisture Macro F1-score	0.3576	0.6546	+0.2970	83.05%
Edge Macro F1-score	0.7403	0.8314	+0.0911	12.31%

The comparison in Table 4.3 showed the final proposed SSL model improved all 3 IME tasks compared with the supervised baseline. The improvement of Moisture task was the largest, where the Macro F1-score increased from 0.3576 to 0.6546, giving a relative improvement of 83.05%. Infection and Edge also improved by 6.83% and 12.31%, respectively. Therefore, the Round 3 Moist_F1 0.70 model was selected as the final proposed model.

4.4 Mobile Prototype Integration

The final proposed model was successfully integrated into a smartphone-based prototype. The deployment workflow used Hugging Face Spaces as the cloud inference endpoint, while the mobile application was developed using Flutter and Dart and tested using the Android Studio emulator.

In the prototype workflow, a wound image is uploaded from the mobile application to the cloud inference endpoint. The image is then processed through the TIME-Net pipeline, which includes tissue segmentation, preprocessing, tissue classification, and IME classification. The prediction results are returned to the Flutter application and displayed as TIME-based wound assessment outputs.



(a) (b)

Figure 4.2: Mobile prototype interface showing (a) tissue breakdown and IME assessment output and (b) previous and current comparison

In the analysis function, the mobile emulator displayed the tissue breakdown result together with the IME assessment. The IME assessment predicted Infection, Moisture, and Edge conditions with their respective confidence scores as shown in Figure 4.2 (a). The prototype also includes a comparison function for follow-up wound monitoring, as shown in Figure 4.2 (b), which compares previous and current IME assessment results side by side.

4.5 Inference Time Evaluation

The inference time was also measured to evaluate the feasibility of the cloud-based workflow. The total server inference time was 2692.18ms, which is approximately 2.69 seconds. The segmentation process required the longest time at 1808.94ms, followed by IME classification 473.14ms and tissue classification at 410.09ms.

Table 4.4: Inference time of the mobile prototype

Processing Stage	Time (ms)	Time (s)
Segmentation	1808.94	1.81
Tissue Classification	410.09	0.41
IME Classification	473.14	0.47
Total Server Inference	**2692.18**	**2.69**

These results show that the proposed TIME-Net workflow can be deployed through a cloud-based inference service and connected to a smartphone application. The prototype was able to return the tissue and IME assessment results within a short response time.

CHAPTER 5: DISCUSSION

5.1 Effect of Segmentation on IME Classification

Based on the results, segmented input improved IME classification by reducing irrelevant background information and allowed the model to focus on the RGB wound-specific features. This is consistent with Anisuzzaman et al. (2021) studies, where segmentation is commonly used to isolate the wound region and reduce background noise before further assessment.

The improvement of Infection and Edge tasks was obvious as these two tasks depend strongly on wound boundary and periwound information, such as redness, tissue transition, and wound margin. The padding added around the wound during preprocessing may have helped retain these visual cues. In comparison with Moisture task, the improvement was smaller. This may be caused by Moisture class having 3 different classes (too_wet, moisture, and too_dry), thus, the model needs to distinguish between more categories and increase the difficulty of classification. Zhang et al. (2021) developed a mobile health tool for home-based wound monitoring and showed that image quality is an important factor for reliable wound assessment. Their findings help justify that moisture results can be changed depending on lighting, reflection, and exudate visibility. This also suggests that although segmentation helps the model focus on the wound area, some IME features are still difficult to classify using image information alone.

5.2 Backbone Architecture Selection

The backbone architecture comparison also supports selecting EfficientNet-B0 as the feature extractor for IME-ClassNet. Morid et al. (2021) reviewed transfer learning studies using ImageNet-pretrained models in medical image analysis and showed that transfer learning is commonly utilized when labelled datasets are limited. This matches the data in this study, which is very limited labelled data available.

Although ResNet50 achieved a higher Moisture Macro F1-score, EfficientNet-B0 achieved the best average Macro F1-score across the 3 IME tasks, while using fewer parameters than ResNet and Swin_t. This supports the selection of EfficientNet-B0 as a balanced backbone for overall performance and deployment efficiency.

5.3 Effect of Semi-Supervised Learning

In Cheplygina et al. (2019) study, researchers reviewed semi-supervised and transfer learning methods in medical image analysis and found that limited annotated data is a common challenge in medical imaging. Liu et al. (2023) applied semi-supervised learning with EfficientNet-B0 for chronic wound image assessment by using a small labelled wound dataset together with a large unlabeled dataset, which has a similar setup as this study. In their research, they also using unlabeled wound images to improve learning under limited labelled subset. In this study, pseudo-labelling also proved to successfully improve all 3 IME tasks compared with the supervised baseline, with the largest improvement observed in Moisture task. This suggests that the additional pseudo-labelled images increased the model training images and helped the model to learn more visual patterns that were not sufficiently represented in the small labelled subset of the public dataset.

5.4 Pseudo-Label Quantity and Quality

Although semi-supervised learning improved model performance, the results also showed that adding more pseudo-labelled data does not always lead to better performance. Round 4 generated more high-confidence pseudo-labels than Round 3, but did not achieve the best Moisture Macro F1-score. The active Moisture pseudo-labeled increased from 530 in Round 3 to 2243 in Round 4, but the best Moisture performance was still achieved by Round 3 model.

This may be due to confirmation bias, where the model is confident in its prediction, but incorrect pseudo-labels are added into training and reinforce existing errors. From the results, Round 4 generated many more high-confidence pseudo-labels, but the highest Moisture Macro F1-score was still achieved by Round 3 model. Therefore, the round 3 model was selected as the final model, which was selected based on the validation Macro F1-score rather than the number of pseudo-labels alone.

5.5 Mobile Prototype Feasibility

The mobile prototype integration showed that the proposed TIME-Net workflow can function as an end-to-end proof-of-concept system. The Flutter application was able to display tissue breakdown, IME classification results, confidence scores, and the comparison of wound progressions. This demonstrates that the model can be connected to a smartphone-based workflow and return TIME-based assessment results to the user.

The total server inference time was 2.69 seconds, which shows that the cloud-based workflow can return results within a short response time. Wang et al. (2021) discussed that mobile deep learning applications need to balance model accuracy, speed, memory, battery usage, and deployment constraints. In this study, Hugging Face Spaces was used as the cloud inference endpoint. This approach reduced the need for direct on-device model conversion and allowed the model to be updated more easily. However, this also means that response time is highly dependent on the user's network speed. Therefore, the current system should be viewed as a prototype that demonstrates technical feasibility, rather than a fully optimized clinical deployment.

5.6 Limitations and Future Work

This study has some limitations, which include the small labelled subset, which may limit the model's generalization across different wound types and capture conditions. Although pseudo-labelling increased the training data, it may still contain errors as it was predicted by the weak model. Second, the IME labels were imbalanced, especially for Moisture task. This limits the model from learning from the minority class, even though class weighting and Macro F1-score were used. Third, the model prediction is mainly on image color and image visible features. However, some wound conditions may not be observable based on an image and would still require a physical wound assessment to confirm their condition. Therefore, the proposed system should be treated as a clinical support tool rather than a replacement for clinical judgement. Future work should include a larger and more diverse dataset, improved pseudo-label filtering strategies, and clinical usability testing. Further deployment work can also explore inference optimization or on-device inference to reduce dependency on cloud connection.

CHAPTER 6: CONCLUSION

6.1 Conclusion

This study successfully developed TIME-Net, an AI-driven wound assessment framework based on the TIME framework. The proposed framework combines T-SegNet for wound-focused image generation and IME-ClassNet for multi-task classification of Infection, Moisture, and Edge conditions. The use of T-SegNet segmentation output helped generate wound-focused images and reduce irrelevant background information before IME classification.

The findings showed that segmented wound-focused images improved IME classification compared with using original raw wound images directly. EfficientNet-B0 was selected as the model backbone architecture because it is lightweight, which is suitable for mobile deployment and achieved the best overall balance between Macro F1-score and model efficiency. The semi-supervised pseudo-labelling approach also further improved performance compared with the supervised baseline, especially for Moisture task, which showed the largest improvement.

Round 3 model was selected as the final proposed model as it has achieved the best final Moisture performance. This selected model was also successfully integrated into a smartphone prototype through HuggingFace Spaces. The total server inference time of approximately 2.69 seconds demonstrated the feasibility of TIME-Net as a proof-of-concept mobile wound assessment workflow.

Overall, this study shows the potential of combining segmentation, multi-task classification, and semi-supervised learning to support more structured wound assessment. Future work should focus on larger labelled datasets, improved pseudo-label filtering, and deployment optimization before real clinical use.

6.2 Contributions of the Study

The main contributions of this study are as follows:

1. This study developed IME-ClassNet, a multi-task deep learning model that classifies Infection, Moisture, and Edge conditions of chronic wounds in a single workflow.
2. This study integrated T-SegNet and IME-ClassNet into the proposed TIME-Net framework, where segmentation output was used to generate wound-focused images before IME classification.
3. This study applied semi-supervised pseudo-labelling to address the limitation of small labelled IME data and demonstrated improvement over the supervised baseline.
4. This study evaluated different backbone architectures and selected EfficientNet-B0 as the final backbone based on its balance between classification performance and model efficiency.
5. This study demonstrated the feasibility of integrating the final TIME-Net workflow into a smartphone-based prototype using a cloud-hosted inference pipeline.

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